SAMAKSHA AGARWAL

Java Backend Developer · Distributed Systems

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SUMMARY

Third-year CS undergraduate (JIIT Noida, GPA 8.82/10) building distributed, event-driven systems in Java 21 and Spring Boot. Hands-on experience with concurrent multi-service architectures and scalable REST API design. 500+ DSA problems solved; strong foundation in algorithms, data structures.

SKILLS

Languages: Java 21, C++, Python
Backend & APIs: Spring Boot 3, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST API Design
Distributed Systems: Apache Kafka, Microservices, Event-Driven Architecture
Databases: MySQL, PostgreSQL
DevOps & Observability: Docker, GitHub Actions, Prometheus, Grafana, Git, Maven, Linux/Shell
Testing & Tools: JUnit 5, Mockito, Postman, IntelliJ IDEA
AI Integration: Gemini AI API, Spring AI
Core CS: Data Structures & Algorithms (Graphs, DP, Greedy), OOP, OS Concepts

EXPERIENCE

Software Engineering Virtual Program *01/2026 · Remote*
JPMorgan Chase & Co.

- Integrated Apache Kafka into a Spring Boot microservice; consumed and deserialized high-volume transaction messages concurrently — processed **10,000+ messages per run with zero data loss**, validating multi-threaded consumer behaviour
- Implemented distributed transaction validation and persistence using Spring Data JPA; applied transactional integrity across relational User records with H2 in-memory database
- Developed REST endpoints for user balance queries and external Incentive API integration via RestTemplate

EDUCATION

B. Tech, Computer Science and Engineering — Jaypee Institute of Information Technology **GPA: 8.82/10.0**
07/2024 – 07/2028 · Noida, India · Relevant: DSA, Design & Analysis of Algorithms, DBMS, OS, Software Engineering
Senior Secondary (Class XII) — St. Peter's School 05/2023 · 93.75/100%
Secondary (Class X) — St. Peter's School 05/2021 · 93.50/100%

PROJECTS

RainIQ – Distributed Harvesting Platform **GitHub** · **Demo** *September 2026*
Java 21 · Spring Boot 3 · Apache Kafka · PostgreSQL · Flyway · Spring Security · JWT · Docker · GitHub Actions · Prometheus/Grafana

- Monitored via Prometheus/Grafana: **sub-300ms P95-range latency, continuous uptime across all 8 services during testing** — measured under local load
- Built a **10-service distributed backend** (Auth, Property, Design, Rainfall, Compliance, AI, Monitoring, Notification, Eureka, API Gateway) with strict **database-per-service isolation**
- Designed an event-driven compliance pipeline via Kafka — design submission triggers async rainfall-based evaluation, fanning out to AI recommendations (Gemini API) on pass or email alerts on fail
- Implemented JWT auth and role-based access control across **5 roles**, independently validated at each service
- Containerized full stack with **Docker Compose**; built **GitHub Actions CI/CD** pushing per-service images to Docker Hub across all 10 services

Weather Dashboard App **GitHub** *March 2026*
Java 21 · Spring Boot · Spring MVC · REST APIs · OpenWeatherMap API

- Built stateless Spring Boot backend consuming OpenWeatherMap REST API; serves real-time conditions and 3-day forecasts — designed for horizontal scalability
- Structured backend following Controller → Service → DTO → Entity layered architecture; all REST endpoints tested and verified via Postman across multiple city inputs including edge cases (invalid city, API timeout handling)

BookMyShow Backend Clone **GitHub** *June 2026*
Java 21 · Spring Boot · Hibernate / JPA · MySQL · Maven

- Designed relational schema (Users, Movies, Theatres, Screens, Seats, Shows, Showseats, Payments, Bookings); applied pessimistic locking on seat reservations to prevent double-booking under concurrent load
- Structured backend with Controller → Service → DTO → Entity → Repository layers across 9 core entities; all booking flow endpoints tested end-to-end via Postman — verified seat selection, booking confirmation, and cancellation flows

KEY ACHIEVEMENTS

LeetCode: **Sk_Agar** — Max Rating 1563 · 500+ problems solved across Arrays, Graphs, Trees, DP, Sliding Window
Multi-platform: Active on **Codeforces** and **CodeChef** — consistent practice in graph algorithms and dynamic programming.
Academic: Top 5 in district · Class XII ISC 93.75% · 2023